

Phishing Email Threat Investigation & Incident Response

Simulated Cybersecurity Incident Report

Analyst: Aliyah Flowers

Incident Type: Phishing / Potential Credential Harvesting

Severity: High

Organization: Northstar Financial Services (fictional)

Simulation Notice

This report documents a controlled cybersecurity simulation created for educational and portfolio purposes. No real organization, employee, credentials, or malicious infrastructure were involved.

1. Executive Summary

A fictional employee received an email claiming to be from Microsoft 365 Security. The message threatened account suspension within 24 hours and directed the recipient to an external account-verification URL. The investigation identified multiple characteristics consistent with phishing, including brand impersonation, urgency, fear-based messaging, a suspicious sender domain, and a potential credential-harvesting destination.

The simulated URL was checked using VirusTotal and returned "Item not found." The simulated domain was checked using ICANN Lookup and was reported as not available. These results were not treated as evidence that the URL was safe.

2. Incident Scenario

Employee: Jordan Williams, Accounts Payable Specialist

Organization: Northstar Financial Services

Claimed Sender: Microsoft 365 Security

Subject: URGENT: Your Microsoft 365 account will be suspended

The message instructed the employee to complete an immediate security verification process to avoid losing access to Microsoft 365 services.

3. Investigation Objectives

- Identify suspicious email characteristics
- Analyze sender and domain information
- Examine the included URL
- Perform basic threat-intelligence checks
- Identify indicators of compromise
- Assess potential impact
- Recommend incident-response actions

4. Technical Analysis

Sender: security@microsoft-365support.com

The sender domain appears designed to resemble a Microsoft-related security address while not representing an official Microsoft domain.

URL: https://microsoft365-security-verification.example/login

The URL uses a Microsoft-themed hostname and a /login path, creating a potential credential-harvesting scenario. The .example domain is reserved for documentation/testing and is used here only as part of the simulation.

Social-engineering indicators: urgency, fear of account suspension, brand impersonation, and a request for immediate account verification.

5. Threat Intelligence Results

VirusTotal: Item not found. No existing detection information was available for the simulated URL.

ICANN Lookup: Not available. The .example domain used in this simulation does not represent a registered production domain.

Analyst note: The absence of threat-intelligence results does not establish that a URL is legitimate. Assessment should consider the complete body of evidence.

6. Indicators of Compromise

Type	Indicator	Assessment
Email	security@microsoft-365support.com	Suspicious sender domain
Domain	microsoft-365support.com	Potential Microsoft impersonation
URL	microsoft365-security-verification.example/login	Suspicious verification URL
URL Path	/login	Potential credential-harvesting destination
Social Engineering	24-hour suspension warning	Urgency tactic
Social Engineering	Account verification request	Potential credential theft

7. Risk Assessment

Overall Risk: High

If a recipient interacted with a real phishing page and submitted credentials, potential consequences could include account compromise, unauthorized access to business information, exposure of email and documents, internal phishing propagation, and further compromise of organizational resources.

Risk-reduction controls include MFA, email filtering, security awareness training, phishing-reporting mechanisms, domain/URL filtering, centralized logging, and monitoring for suspicious authentication activity.

8. Incident Response Recommendations

1. Quarantine and remove the phishing message.
2. Search for additional copies in other mailboxes.
3. Block confirmed malicious indicators where appropriate.
4. Determine whether the suspicious URL was accessed.
5. Determine whether credentials were submitted.
6. Reset potentially compromised credentials.

7. Revoke active sessions or authentication tokens when appropriate.
8. Verify MFA is enabled and review authentication methods.
9. Review account and authentication activity.
10. Notify affected employees.
11. Document findings and lessons learned.

9. Conclusion

The simulated email was assessed as a phishing attempt based on multiple independent characteristics: Microsoft brand impersonation, a suspicious sender domain, urgency, fear-based messaging, and an external account-verification URL. The investigation demonstrates an evidence-based workflow in which individual indicators are considered together rather than relying on a single threat-intelligence result.

10. Skills Demonstrated

Phishing analysis • Social-engineering analysis • Threat intelligence • IOC identification • Risk assessment • Incident response • Security documentation • URL/domain analysis • GitHub • Markdown

Portfolio Project — Aliyah Flowers | Entry-Level Cybersecurity Professional